Bitcoin Transaction Experiment: Legacy (P2PKH) and SegWit (P2SH-P2WPKH) Implementation

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1. OBJECTIVE:

The objective of this lab is to understand how Bitcoin transactions work by programmatically creating and validating transactions using Legacy (P2PKH) and SegWit (P2SH-P2WPKH) address formats.

The experiment was performed on the Bitcoin Regtest network, which allows instant mining and local testing without real Bitcoin.

Regtest provides a controlled environment where transactions, scripts, and blocks can be created and analyzed easily.

2. Environment Setup :

- Bitcoin Core was run in **regtest mode**
- RPC connection was established using Python
- Wallet was created/loaded
- 101 blocks were mined to obtain spendable coins

Bitcoin Core daemon (bitcoind) was started in regtest mode.
A wallet named **legacywall** was created and connected through RPC using Python.
101 blocks were mined using generatetoaddress so that the first block reward becomes spendable.

3. TASK 1 : Legacy Transaction(P2PKH)

Three legacy addresses were generated.

A: mn8Z9m4CV64Gmz98fYet1gS3iSvVy6NzpG
B: misYrLkDWUihDTYfghY33bwPoe58549SwM
C: n2PbR3yPPskP6pxjibxNy2YwkYnYRERiUa

These addresses follow the **P2PKH format**, which locks coins using the hash of a public key.

Funding Address A

Transaction: sendtoaddress A 10 BTC

TXID: d6bc10eee00ddf4efcbababe1507d0946dcb85005943fcc5daad590ee6901b3d

Address A was funded with 10 BTC. This transaction created a UTXO which will be used as input for the next transaction.

Transaction A → B

Input:

- UTXO from funding transaction

Outputs:

- B: 5 BTC
- A: 4.9999 BTC (change)

Transaction fee: 0.0001 BTC

TXID:	43ce07a410238400d1bfd60d06fa1d1f99d03764675f6e948017894d77f3a2af
scriptPubKey.asm:	OP_DUP OP_HASH160 24cd8b052ee1db583d0028a23edd5484a57b7794 OP_EQUALVERIFY OP_CHECKSIG
scriptPubKey.type:	pubkeyhash
scriptSig.asm:	304402203c59441c4973f438373e56a4c23f3b87433a85b4559aa9a862ddbcb62cd474e90220476b1f5470025bd60f495855f66da9d728d28ecd9eeaa95c87f68ae4ff84b092[ALL]

0308bdf27f65eed01af2ac17c4d842b0d1c3c16262ab4ade5efe82e72c30902cd3

Size: 225 bytes

Vsize: 225 vbytes

Weight: 900 WU

```
Connected to Bitcoin regtest
Wallet loaded

Generated Addresses
A: mn8Z9m4CV64Gmz98fYet1gS3iSvVy6NzpG
B: misYrLkDWUihDTYfghY33bwPoe58549SwM
C: n2PbR3yPPskP6pxjibxNy2YwkYnYRERiUa
Balance after mining: 50.00000000

Funding transaction to A: d6bc10eee00ddf4efcbababe1507d0946dcb85005943fcc5daad590ee6901b3d

UTXO for A
{'txid': 'd6bc10eee00ddf4efcbababe1507d0946dcb85005943fcc5daad590ee6901b3d', 'vout': 1, 'address': 'mn8Z9m4CV64Gmz98fYet1gS3iSvVy6NzpG', 'label': '', 'scriptPubKey': '76a914488c69186b2d566389726fdca184
cbebd8332a2388ac', 'amount': Decimal('10.00000000'), 'confirmations': 1, 'spendable': True, 'solvable': True, 'desc': 'pkh([c989a8ec/44h/1h/0h/0/3]0308bdf27f65eed01af2ac17c4d842b0d1c3c16262ab4ade5efe82
e72c30902cd3)#73j9nf30', 'parent_descs': ['pkh([c989a8ec/44h/1h/0h]tpubDCJ8NmupjoCEZ4GUgGEsPSr9qnhRyVtvEoTdBDBS3YydYtsRSo1LrTuZ8wEDg9c7gAUKFicyVXTcHroXZFhhn3FbtgR7epm7ZAobcBRa874/0/*)#smede79t'], 'safe
': True}

Raw Transaction A->B
02000000013d1b90e60e59addac5fc43590085cb6d94d00715bebabafc4edf0de0ee10bcd6010000000fdffffff020065cd1d000000001976a91424cd8b052ee1db583d0028a23edd5484a57b779488acf03dcd1d000000001976a914488c69186b2d566
389726fdca184cbebd8332a2388ac00000000

Signed Transaction HEX
02000000013d1b90e60e59addac5fc43590085cb6d94d00715bebabafc4edf0de0ee10bcd6010000006a47304402203c59441c4973f438373e56a4c23f3b87433a85b4559aa9a862ddbcb62cd474e90220476b1f5470025bd60f495855f66da9d728d28ec
d9eeaa95c87f68ae4ff84b09201210308bdf27f65eed01af2ac17c4d842b0d1c3c16262ab4ade5efe82e72c30902cd3fdffffff020065cd1d000000001976a91424cd8b052ee1db583d0028a23edd5484a57b779488acf03dcd1d000000001976a914488c
69186b2d566389726fdca184cbebd8332a2388ac00000000

Decoded Transaction A->B
{'txid': '43ce07a410238400d1bfd60d06fa1d1f99d03764675f6e948017894d77f3a2af', 'hash': '43ce07a410238400d1bfd60d06fa1d1f99d03764675f6e948017894d77f3a2af', 'version': 2, 'size': 225, 'vsize': 225, 'weight
': 900, 'locktime': 0, 'vin': [{'txid': 'd6bc10eee00ddf4efcbababe1507d0946dcb85005943fcc5daad590ee6901b3d', 'vout': 1, 'scriptSig': {'asm': '304402203c59441c4973f438373e56a4c23f3b87433a85b4559aa9a862dd
bcb62cd474e90220476b1f5470025bd60f495855f66da9d728d28ecd9eeaa95c87f68ae4ff84b092[ALL] 0308bdf27f65eed01af2ac17c4d842b0d1c3c16262ab4ade5efe82e72c30902cd3', 'hex': '47304402203c59441c4973f438373e56a4c23f
3b87433a85b4559aa9a862ddbcb62cd474e90220476b1f5470025bd60f495855f66da9d728d28ecd9eeaa95c87f68ae4ff84b09201210308bdf27f65eed01af2ac17c4d842b0d1c3c16262ab4ade5efe82e72c30902cd3'}, 'sequence': 4294967293}
], 'vout': [{'value': Decimal('5.00000000'), 'n': 0, 'scriptPubKey': {'asm': 'OP_DUP OP_HASH160 24cd8b052ee1db583d0028a23edd5484a57b7794 OP_EQUALVERIFY OP_CHECKSIG', 'desc': 'addr(misYrLkDWUihDTYfghY33
bwPoe58549SwM)#m5zd6u6t', 'hex': '76a91424cd8b052ee1db583d0028a23edd5484a57b779488ac', 'address': 'misYrLkDWUihDTYfghY33bwPoe58549SwM', 'type': 'pubkeyhash'}}, {'value': Decimal('4.99990000'), 'n': 1,
'scriptPubKey': {'asm': 'OP_DUP OP_HASH160 488c69186b2d566389726fdca184cbebd8332a23 OP_EQUALVERIFY OP_CHECKSIG', 'desc': 'addr(mn8Z9m4CV64Gmz98fYet1gS3iSvVy6NzpG)#vzv8rj4d', 'hex': '76a914488c69186b2d5
66389726fdca184cbebd8332a2388ac', 'address': 'mn8Z9m4CV64Gmz98fYet1gS3iSvVy6NzpG', 'type': 'pubkeyhash'}}]}

Broadcasted TXID A->B: 43ce07a410238400d1bfd60d06fa1d1f99d03764675f6e948017894d77f3a2af
```

Transaction B → C

Inputs:

- UTXO from previous transaction was used.

The output sent to address B in the previous transaction becomes the **UTXO input** for this transaction.

Outputs:

- C: 4 BTC
- B: 0.9999 BTC (change)

Transaction Fee: 0.0001 BTC

TXID: 61d545bb6facf91ff42ccd6910e0042ac42d61694b793fbb3d1b21cb51cec2ad

scriptPubKey.asm:	OP_DUP OP_HASH160 e4f64cef67f6a5f548a81ba85ff8a729f3a26c75 OP_EQUALVERIFY OP_CHECKSIG
scriptPubKey.type:	pubkeyhash
scriptSig.asm:	30440220044254019c5b3c0522540f1dd173f85df0d69cbf1d0c1b0293fe78ef212938c902200679bb9e9fc2c34b1b8da1470b7c4221b50bba01ff2f343b115a141ed27f304d[ALL] 03140034273813766a8fb1691434cf6bdeeebef8b2503c804feff130495d78d93c
Size:	225 bytes
Vsize:	225 vbytes
Weight:	900 WU

```
Broadcasted TXID A->B: 43ce07a410238400d1bfd60d06fa1d1f99d03764675f6e948017894d77f3a2af

UTXO for B
{'txid': '43ce07a410238400d1bfd60d06fa1d1f99d03764675f6e948017894d77f3a2af', 'vout': 0, 'address': 'misYrLkDWUihDTYfghY33bwPoe58549SwM', 'label': '', 'scriptPubKey': '76a91424cd8b052ee1db583d0028a23edd
5484a57b779488ac', 'amount': Decimal('5.00000000'), 'confirmations': 1, 'spendable': True, 'solvable': True, 'desc': 'pkh([c989a8ec/44h/1h/0h/0/4]03140034273813766a8fb1691434cf6bdeeebef8b2503c804feff13
0495d78d93c)#mnee6kwy', 'parent_descs': ['pkh([c989a8ec/44h/1h/0h]tpubDCJ8NmupjoCEZ46UgGEsPSr9qnhRyVtvEoTdBDBS3YydYtsRSo1LrTuZ8wEDg9c7gAUKFicyVXTcHroXZFhhn3FbtgR7epm7ZAobcBRa874/0/*)#smede79t'], 'safe'
: True}

Raw Transaction B->C
0200000001afa2f3774d891780946e5f676437d0991f1dfa060dd6bfd100842310a407ce430000000000fdffffff020084d71700000000001976a914e4f64cef67f6a5f548a81ba85ff8a729f3a26c7588acf0b9f505000000001976a91424cd8b052ee1db5
83d0028a23edd5484a57b779488ac00000000

Signed Transaction B->C HEX
0200000001afa2f3774d891780946e5f676437d0991f1dfa060dd6bfd100842310a407ce430000000006a47304402200442540 19c5b3c0522540f1dd173f85df0d69cbf1d0c1b0293fe78ef212938c902200679bb9e9fc2c34b1b8da1470b7c4221b50bba0
1ff2f343b115a141ed27f304d0121031400342738137 66a8fb1691434cf6bdeeebef8b2503c804feff130495d78d93cfdffffff020084d71700000000001976a914e4f64cef67f6a5f548a81ba85ff8a729f3a26c7588acf0b9f505000000001976a91424cd
8b052ee1db583d0028a23edd5484a57b779488ac00000000

Decoded Transaction B->C
{'txid': '61d545bb6facf91ff42ccd6910e0042ac42d61694b793fbb3d1b21cb51cec2ad', 'hash': '61d545bb6facf91ff42ccd6910e0042ac42d61694b793fbb3d1b21cb51cec2ad', 'version': 2, 'size': 225, 'vsize': 225, 'weight
': 900, 'locktime': 0, 'vin': [{'txid': '43ce07a410238400d1bfd60d06fa1d1f99d03764675f6e948017894d77f3a2af', 'vout': 0, 'scriptSig': {'asm': '30440220044254019c5b3c0522540f1dd173f85df0d69cbf1d0c1b0293fe
78ef212938c902200679bb9e9fc2c34b1b8da1470b7c4221b50bba01ff2f343b115a141ed27f304d[ALL] 03140034273813766a8fb1691434cf6bdeeebef8b2503c804feff130495d78d93c', 'hex': '47304402200442540 19c5b3c0522540f1dd173
f85df0d69cbf1d0c1b0293fe78ef212938c902200679bb9e9fc2c34b1b8da1470b7c4221b50bba01ff2f343b115a141ed27f304d0121031400342738137 66a8fb1691434cf6bdeeebef8b2503c804feff130495d78d93c'}, 'sequence': 4294967293}
], 'vout': [{'value': Decimal('4.00000000'), 'n': 0, 'scriptPubKey': {'asm': 'OP_DUP OP_HASH160 e4f64cef67f6a5f548a81ba85ff8a729f3a26c75 OP_EQUALVERIFY OP_CHECKSIG', 'desc': 'addr(n2PbR3yPPskP6pxjibxNy
2YWkYnYRERiUa)#pkxwz9mg', 'hex': '76a914e4f64cef67f6a5f548a81ba85ff8a729f3a26c7588ac', 'address': 'n2PbR3yPPskP6pxjibxNy2YWkYnYRERiUa', 'type': 'pubkeyhash'}}, {'value': Decimal('0.99990000'), 'n': 1,
'scriptPubKey': {'asm': 'OP_DUP OP_HASH160 24cd8b052ee1db583d0028a23edd5484a57b7794 OP_EQUALVERIFY OP_CHECKSIG', 'desc': 'addr(misYrLkDWUihDTYfghY33bwPoe58549SwM)#m5zd6u6t', 'hex': '76a91424cd8b052ee1d
b583d0028a23edd5484a57b779488ac', 'address': 'misYrLkDWUihDTYfghY33bwPoe58549SwM', 'type': 'pubkeyhash'}}]}

Broadcasted TXID B->C: 61d545bb6facf91ff42ccd6910e0042ac42d61694b793fbb3d1b21cb51cec2ad

Final Balance: 199.99978530
```

Script Analysis (P2PKH):

Locking script (scriptPubKey):

OP_DUP
OP_HASH160
<pubKeyHash>
OP_EQUALVERIFY
OP_CHECKSIG

Unlocking script (scriptSig):

<signature>
<publicKey>

Execution:

1. Public key is hashed
2. Compared with pubKeyHash
3. Signature verified using OP_CHECKSIG

If valid → stack returns TRUE → transaction accepted.

Bitcoin Script Debugging (BTCDEB):

```
┌──(sumitmodanwal㉿kali)-[~]
└─$ btcdeb 76a91424cd8b052ee1db583d0028a23edd5484a57b779488ac 3044022004425401 9c5b3c0522540f1dd173f85df0d69cbf1d0c1b0293fe78ef212938c902200679bb9e9fc2c34b1b8
da1470b7c4221b50bba01ff2f343b115a141ed27f304d01 03140034273813766a8fb1691434cf6bdeeebef8b2503c804feff130495d78d93c
btcdeb 5.0.24 -- type `btcdeb -h` for start up options
LOG: signing segwit taproot
notice: btcdeb has gotten quieter; use --verbose if necessary (this message is temporary)
5 op script loaded. type `help` for usage information
script                                  |                                                                  stack
----------------------------------------+------------------------------------------------------------------
OP_DUP                                  | 03140034273813766a8fb1691434cf6bdeeebef8b2503c804feff130495d78d93c
OP_HASH160                              | 3044022004425401 9c5b3c0522540f1dd173f85df0d69cbf1d0c1b0293fe78e...
24cd8b052ee1db583d0028a23edd5484a57b7794 |
OP_EQUALVERIFY                          |
OP_CHECKSIG                             |
#0000 OP_DUP
btcdeb> step
                <> PUSH stack 03140034273813766a8fb1691434cf6bdeeebef8b2503c804feff130495d78d93c
script                                  |                                                                  stack
----------------------------------------+------------------------------------------------------------------
OP_HASH160                              | 03140034273813766a8fb1691434cf6bdeeebef8b2503c804feff130495d78d93c
24cd8b052ee1db583d0028a23edd5484a57b7794 | 03140034273813766a8fb1691434cf6bdeeebef8b2503c804feff130495d78d93c
OP_EQUALVERIFY                          | 3044022004425401 9c5b3c0522540f1dd173f85df0d69cbf1d0c1b0293fe78e...
OP_CHECKSIG                             |
#0001 OP_HASH160
btcdeb>
                <> POP  stack
                <> PUSH stack 24cd8b052ee1db583d0028a23edd5484a57b7794
script                                  |                                                                  stack
----------------------------------------+------------------------------------------------------------------
```

```
----------------------------------------+------------------------------------------------------------------
24cd8b052ee1db583d0028a23edd5484a57b7794 |                                 24cd8b052ee1db583d0028a23edd5484a57b7794
OP_EQUALVERIFY                          | 03140034273813766a8fb1691434cf6bdeeebef8b2503c804feff130495d78d93c
OP_CHECKSIG                             | 3044022004425401 9c5b3c0522540f1dd173f85df0d69cbf1d0c1b0293fe78e...
#0002 24cd8b052ee1db583d0028a23edd5484a57b7794
btcdeb>
                <> PUSH stack 24cd8b052ee1db583d0028a23edd5484a57b7794
script                                  |                                                                  stack
----------------------------------------+------------------------------------------------------------------
OP_EQUALVERIFY                          |                                 24cd8b052ee1db583d0028a23edd5484a57b7794
OP_CHECKSIG                             |                                 24cd8b052ee1db583d0028a23edd5484a57b7794
                                        | 03140034273813766a8fb1691434cf6bdeeebef8b2503c804feff130495d78d93c
                                        | 3044022004425401 9c5b3c0522540f1dd173f85df0d69cbf1d0c1b0293fe78e...
#0003 OP_EQUALVERIFY
btcdeb>
                <> POP  stack
                <> POP  stack
                <> PUSH stack 01
                <> POP  stack
script                                  |                                                                  stack
----------------------------------------+------------------------------------------------------------------
```

```
----------------------------------------+-----------------------------------------------------------------
OP_CHECKSIG                             | 03140034273813766a8fb1691434cf6bdeeebef8b2503c804feff130495d78d93c
                                        | 30440220044254019c5b3c0522540f1dd173f85df0d69cbf1d0c1b0293fe78e...
#0004 OP_CHECKSIG
btcdeb>
EvalChecksig() sigversion=0
Eval Checksig Pre-Tapscript
error: Signature must be zero for failed CHECK(MULTI)SIG operation
btcdeb>
```

Note : OP_EQUALVERIFY successfully matched the generated public key hash with the destination address hash. The final OP_CHECKSIG operation intentionally fails in this standalone btcdeb environment because the debugger lacks the full transaction context (inputs, outputs, amounts) required to independently calculate the transaction hash and verify the ECDSA signature mathematically.

If the user **continues stepping after the script has already completed**, btcdeb will simply report: **at end of script**

This indicates that **no more instructions remain to be executed in the script**. The debugger has reached the final instruction boundary, and any further step commands do not correspond to executable opcodes.

The locking script (scriptPubKey) follows the P2PKH format:
OP_DUP OP_HASH160 <pubKeyHash> OP_EQUALVERIFY OP_CHECKSIG.
The unlocking script (scriptSig) contains the signature and public key required to validate ownership. During transaction validation, the public key hash is compared with the hash in the locking script and the signature is verified using OP_CHECKSIG.

Execution Flow:

push signature

push pubkey

duplicate pubkey

hash160(pubkey)

compare with pubKeyHash

verify signature

4. TASK 2 : P2SH-SegWit Transactions

SegWit Address Generation

Generated addresses:

A’ : 2MtUsWqe3zjMZNpCgNo6zurytSbMs8gb6W2
B': 2NCntr9UK6YrY51Y18RJGq2RK4FBXi7p9Nb
C': 2N7kLRW5YbUDq1bLpEvbPpJ3XJr47YPtri2

These are **P2SH-wrapped SegWit addresses**, where the witness data is separated from the main transaction.

Funding A'

TXID: 39839b3189b5ab9bddf6ef5e14f585905ba0f77d2f7c2fa3197cb122d1375b29

Created a SegWit UTXO.

Transaction A' → B'

Inputs:

- UTXO created by previous fund transaction(10 BTC).

Outputs:

- B': 5 BTC
- A': 4.9999 BTC

Transaction Fee: 0.0001 BTC

TXID:	fdfe2cacd43ddb287527ebb9b835b7066c50d860268aac2052ee50e28d23e4c7
scriptPubKey.asm:	OP_HASH160 d667b920bc00607f042d93dcf3fa127abaf05284 OP_EQUAL
scriptPubKey.type:	scripthash
scriptSig.asm:	001463d1587df1ea99d0177addb66fa328ad7949093a
Size:	247 bytes
Vsize:	166 vbytes
Weight:	661 WU

```
Connected to Bitcoin regtest
Wallet loaded

Generated SegWit Addresses
A': 2MtUsWqe3zjMZNpCgNo6zurytSbMs8gb6W2
B': 2NCntr9UK6YrY51Y18RJGq2RK4FBXi7p9Nb
C': 2N7kLRW5YbUDq1bLpEvbPpJ3XJr47YPtri2
Balance after mining: 50.00000000

Balance after mining: 50.00000000

Funding transaction to A': 39839b3189b5ab9bddf6ef5e14f585905ba0f77d2f7c2fa3197cb122d1375b29

UTXO for A'
{'txid': '39839b3189b5ab9bddf6ef5e14f585905ba0f77d2f7c2fa3197cb122d1375b29', 'vout': 1, 'address': '2MtUsWqe3zjMZNpCgNo6zurytSbMs8gb6W2', 'label': '', 'redeemScript': '001463d1587df1ea99d0177addb66fa32
8ad7949093a', 'scriptPubKey': 'a9140d8cdbedcf6b2246bb5585a8bbece79b84f9171087', 'amount': Decimal('10.00000000'), 'confirmations': 1, 'spendable': True, 'solvable': True, 'desc': 'sh(wpkh([c50794bc/49h
/1h/0h/0/3]03435daee22c52a384634a6fd2843b76ec44584ae7991d63e643ffd5955cacf6e3))#ncyd60cu', 'parent_descs': ['sh(wpkh([c50794bc/49h/1h/0h]tpubDDW8hrcBwrxNVAcfyB9Xak43K3NGjmVybsq9Aw8J5mLkmuHcKnc5crf3D3Xd
4AARJgRhGguiXa2P14hGn5ffJEuBCFaCvw9P1srbJeH4K4q/0/*))#vq6jluhr'], 'safe': True}

Raw Transaction A'->B'
0200000001295b37d122b17c19a32f7c2f7df7a05b9085f5145eeff6dd9babb589319b83390100000000fdffffff020065cd1d0000000017a914d667b920bc00607f042d93dcf3fa127abaf0528487f03dcd1d0000000017a9140d8cdbedcf6b2246bb558
5a8bbece79b84f91710870000000000

Signed Transaction HEX
020000000000101295b37d122b17c19a32f7c2f7df7a05b9085f5145eeff6dd9babb589319b83390100000001716001463d1587df1ea99d0177addb66fa328ad7949093afdffffff020065cd1d0000000017a914d667b920bc00607f042d93dcf3fa127abaf
0528487f03dcd1d0000000017a9140d8cdbedcf6b2246bb5585a8bbece79b84f91710870247304402205d213450e6171f5b0cd5a5bb9f79b65114051f3fd6a376298385c6965f6853a02205b8cf8b15815437e9fcb1b21ffca6d5c61240c0e9f1dca8384
162029d069c1460121034 35daee22c52a384634a6fd2843b76ec44584ae7991d63e643ffd5955cacf6e300000000

Decoded Transaction A'->B'
{'txid': 'fdfe2cacd43ddb287527ebb9b835b7066c50d860268aac2052ee50e28d23e4c7', 'hash': '381c82ed696ec6a6fb168a642673803a12446faf68e121d33334fda3f658f6a0', 'version': 2, 'size': 247, 'vsize': 166, 'weight
': 661, 'locktime': 0, 'vin': [{'txid': '39839b3189b5ab9bddf6ef5e14f585905ba0f77d2f7c2fa3197cb122d1375b29', 'vout': 1, 'scriptSig': {'asm': '001463d1587df1ea99d0177addb66fa328ad7949093a', 'hex': '16001
463d1587df1ea99d0177addb66fa328ad7949093a'}, 'txinwitness': ['3044022058d213450e6171f5b0cd5a5bb9f79b65114051f3fd6a376298385c6965f6853a02205b8cf8b15815437e9fcb1b21ffca6d5c61240c0e9f1dca8384162029d069c14
601', '03435daee22c52a384634a6fd2843b76ec44584ae7991d63e643ffd5955cacf6e3'], 'sequence': 4294967293}], 'vout': [{'value': Decimal('5.00000000'), 'n': 0, 'scriptPubKey': {'asm': 'OP_HASH160 d667b920bc00
607f042d93dcf3fa127abaf0528 4 OP_EQUAL', 'desc': 'addr(2NCntr9UK6YrY51Y18RJGq2RK4FBXi7p9Nb)#lne6pgcz', 'hex': 'a914d667b920bc00607f042d93dcf3fa127abaf0528487', 'address': '2NCntr9UK6YrY51Y18RJGq2RK4FBXi
7p9Nb', 'type': 'scripthash'}}, {'value': Decimal('4.99990000'), 'n': 1, 'scriptPubKey': {'asm': 'OP_HASH160 0d8cdbedcf6b2246bb5585a8bbece79b84f91710 OP_EQUAL', 'desc': 'addr(2MtUsWqe3zjMZNpCgNo6zurytS
bMs8gb6W2)#mpc768u7', 'hex': 'a9140d8cdbedcf6b2246bb5585a8bbece79b84f9171087', 'address': '2MtUsWqe3zjMZNpCgNo6zurytSbMs8gb6W2', 'type': 'scripthash'}}]}

Broadcasted TXID A'->B': fdfe2cacd43ddb287527ebb9b835b7066c50d860268aac2052ee50e28d23e4c7
```

Transaction B' → C'

Inputs:

- UTXO created by previous transaction.

Outputs:

- C': 4 BTC
- B': 0.9999 BTC

Transaction Fee: 0.0001 BTC

TXID:	d57ae66fc8db1ea09be230731e401e92e375960fa2c043461224612b891f7621
scriptPubKey.asm:	OP_HASH160 9f1332034964fe8b5ab84cb1dd2b8fb8edbd5c9a OP_EQUAL
scriptPubKey.type:	scripthash
scriptSig.asm:	0014ebaccf2570b37fff170e92758facbf9febca3e18
Size:	247
Vsize:	166
Weight:	661

```
Broadcasted TXID A'->B': fdfe2cacd43ddb287527ebb9b835b7066c50d860268aac2052ee50e28d23e4c7

UTXO for B'
{'txid': 'fdfe2cacd43ddb287527ebb9b835b7066c50d860268aac2052ee50e28d23e4c7', 'vout': 0, 'address': '2NCntr9UK6YrY51Y18RJGq2RK4FBXi7p9Nb', 'label': '', 'redeemScript': '0014ebaccf2570b37fff170e92758facb
f9febca3e18', 'scriptPubKey': 'a914d667b920bc00607f042d93dcf3fa127abaf0528487', 'amount': Decimal('5.00000000'), 'confirmations': 1, 'spendable': True, 'solvable': True, 'desc': 'sh(wpkh([c50794bc/49h/
1h/0h/0/4]032c2601d71dfccee754989c83d7ed763aef7784028eb742e83564d602f260087c))#4rtcyzpx', 'parent_descs': ['sh(wpkh([c50794bc/49h/1h/0h]tpubDDW8hrcBwrxNVAcfyB9Xak43K3NGjmVybsq9Aw8J5mLkmuHCKnc5crf3D3XD4
AARJgRh6guiXa2P14hGn5ffJEuBCFaCvw9P1srbJeH4K4q/0/*))#vq6jluhr'], 'safe': True}

Raw Transaction B'->C'
0200000001c7e4238de250ee5220ac8a2660d8506c06b735b8b9eb277528db3dd4ac2cfefd0000000000fdffffff020084d7170000000017a9149f1332034964fe8b5ab84cb1dd2b8fb8edbd5c9a87f0b9f5050000000017a914d667b920bc00607f042d9
3dcf3fa127abaf052848700000000

Signed Transaction HEX
0200000000101c7e4238de250ee5220ac8a2660d8506c06b735b8b9eb277528db3dd4ac2cfefd0000000017160014ebaccf2570b37fff170e92758facbf9febca3e18fdffffff020084d7170000000017a9149f1332034964fe8b5ab84cb1dd2b8fb8edb
d5c9a87f0b9f5050000000017a914d667b920bc00607f042d93dcf3fa127abaf0528487024730440220193417190432495 48bf5f52c81381e3a677d47b1173664887f2692579869e2fe022070368f33ecbd356eba8318b07a22eaebb71009bc8ff08a4793
972d80f0fda17b0121032c2601d71dfccee754989c83d7ed763aef7784028eb742e83564d602f260087c00000000

Decoded Transaction B'->C'
{'txid': 'd57ae66fc8db1ea09be230731e401e92e375960fa2c043461224612b891f7621', 'hash': '5ff033d234a3dc060c677cbd283c0477c3601654597788cf2288172f2617273d', 'version': 2, 'size': 247, 'vsize': 166, 'weight
': 661, 'locktime': 0, 'vin': [{'txid': 'fdfe2cacd43ddb287527ebb9b835b7066c50d860268aac2052ee50e28d23e4c7', 'vout': 0, 'scriptSig': {'asm': '0014ebaccf2570b37fff170e92758facbf9febca3e18', 'hex': '16001
4ebaccf2570b37fff170e92758facbf9febca3e18'}, 'txinwitness': ['3044022019341719043249548bf5f52c81381e3a677d47b1173664887f2692579869e2fe022070368f33ecbd356eba8318b07a22eaebb71009bc8ff08a4793972d80f0fda17
b01', '032c2601d71dfccee754989c83d7ed763aef7784028eb742e83564d602f260087c'], 'sequence': 4294967293}], 'vout': [{'value': Decimal('4.00000000'), 'n': 0, 'scriptPubKey': {'asm': 'OP_HASH160 9f1332034964
fe8b5ab84cb1dd2b8fb8edbd5c9a OP_EQUAL', 'desc': 'addr(2N7kLRW5YbUDq1bLpEvbPpJ3XJr47YPtri2)#yrtld02u', 'hex': 'a9149f1332034964fe8b5ab84cb1dd2b8fb8edbd5c9a87', 'address': '2N7kLRW5YbUDq1bLpEvbPpJ3XJr47Y
Ptri2', 'type': 'scripthash'}}, {'value': Decimal('0.99990000'), 'n': 1, 'scriptPubKey': {'asm': 'OP_HASH160 d667b920bc00607f042d93dcf3fa127abaf05284 OP_EQUAL', 'desc': 'addr(2NCntr9UK6YrY51Y18RJGq2RK4
FBXi7p9Nb)#lne6pgcz', 'hex': 'a914d667b920bc00607f042d93dcf3fa127abaf0528487', 'address': '2NCntr9UK6YrY51Y18RJGq2RK4FBXi7p9Nb', 'type': 'scripthash'}}]}

Broadcasted TXID B'->C': d57ae66fc8db1ea09be230731e401e92e375960fa2c043461224612b891f7621

Final Balance: 199.99978570
```

SegWit Script Structure:

Locking script:

OP_HASH160 <scriptHash> OP_EQUAL

Redeem script:

0 <pubKeyHash>

Witness data:

signature
public key

Signature data is stored in witness instead of scriptSig.

txinwitness: [‘3044022058d213450e6171f5...’
‘03435daee22c52a384634a6fd2843b76ec44584ae7991d63e643ffd5955cacf6e3’]

First element → ECDSA signature

Second element → Public key

Bitcoin Script Debugging (btcdeb):

```
┌──(sumitmodanwal㉿kali)-[~]
└─$ btcdeb a914d667b920bc00607f042d93dcf3fa127abaf0528487 0014ebaccf2570b37fff170e92758facbf9febca3e18
btcdeb 5.0.24 -- type `btcdeb -h` for start up options
LOG: signing segwit taproot
notice: btcdeb has gotten quieter; use --verbose if necessary (this message is temporary)
6 op script loaded. type `help` for usage information
script                                    |                                          stack
------------------------------------------+------------------------------------------
OP_HASH160                                | 0014ebaccf2570b37fff170e92758facbf9febca3e18
d667b920bc00607f042d93dcf3fa127abaf05284 |
OP_EQUAL                                  |
<<< P2SH script >>>                       |
0                                         |
ebaccf2570b37fff170e92758facbf9febca3e18 |
#0000 OP_HASH160
btcdeb> step
                <> POP  stack
                <> PUSH stack d667b920bc00607f042d93dcf3fa127abaf05284
script                                    |                                          stack
------------------------------------------+------------------------------------------
```

```
------------------------------------------+------------------------------------------
d667b920bc00607f042d93dcf3fa127abaf05284 |     d667b920bc00607f042d93dcf3fa127abaf05284
OP_EQUAL                                  |
<<< P2SH script >>>                       |
0                                         |
ebaccf2570b37fff170e92758facbf9febca3e18 |
#0001 d667b920bc00607f042d93dcf3fa127abaf05284
btcdeb> step
                <> PUSH stack d667b920bc00607f042d93dcf3fa127abaf05284
script                                    |                                          stack
------------------------------------------+------------------------------------------
OP_EQUAL                                  |     d667b920bc00607f042d93dcf3fa127abaf05284
<<< P2SH script >>>                       |     d667b920bc00607f042d93dcf3fa127abaf05284
0                                         |
ebaccf2570b37fff170e92758facbf9febca3e18 |
#0002 OP_EQUAL
btcdeb> step
                <> POP  stack
                <> POP  stack
                <> PUSH stack 01
script                                    |                                          stack
------------------------------------------+------------------------------------------
<<< P2SH script >>>                       |                                            01
0                                         |
ebaccf2570b37fff170e92758facbf9febca3e18 |
<<< P2SH script >>>
btcdeb>
```

In the P2SH-P2WPKH transaction, the locking script (scriptPubKey) uses the format:

OP_HASH160 <redeemScriptHash> OP_EQUAL

The scriptSig contains the **redeem script**, which is the SegWit witness program:

0 <pubKeyHash>

The actual **signature and public key are stored in the witness field (txinwitness) rather than in scriptSig**.

Because witness data is discounted when calculating transaction weight, SegWit transactions have a smaller virtual size and lower transaction fees compared to legacy P2PKH transactions.

Feature	Legacy P2PKH	SegWit P2SH-P2WPKH
scriptSig	contains signature + pubkey	contains redeem script
witness	not used	contains signature + pubkey
btcdeb input	use scriptSig	use witness stack

For legacy transactions, btcdeb directly executes the scriptSig and scriptPubKey because the unlocking data (signature and public key) is stored inside scriptSig.

In SegWit transactions, the signature and public key are stored in the witness field instead of scriptSig.

Therefore, btcdeb must be executed using the witness stack along with the redeem script to validate the transaction.

5. Comparison Table :

Transaction	Size (bytes)	Vsize (vbytes)	Weight (WU)
Legacy A → B	225	225	900
Legacy B → C	225	225	900
Average Legacy	225	225	900
SegWit A' → B'	247	166	661
SegWit B' → C'	247	166	661
Average SegWit	247	166	661

Observation:

- Legacy transactions have **vsize equal to actual size** because all data is counted fully.

- SegWit transactions have **smaller virtual size** because witness data is discounted.

6. Fee Calculation :

Transaction fees depend on **virtual size (vsize)**.

Legacy transaction virtual size : 225 vbytes

SegWit transaction virtual size : 166 vbytes

Percentage reduction:

Savings = (1 − SegWit_vsize / Legacy_vsize) × 100

Savings = (1 − 166 / 225) × 100

Savings ≈ 26.2%

Result:

SegWit transactions reduce the effective transaction size by approximately **26%**, which leads to lower transaction fees.

7. Script Structure Comparison :

The script structure of Legacy P2PKH transactions and SegWit P2SH-P2WPKH transactions differs primarily in how the unlocking data (signature and public key) is stored and processed.

Feature	Legacy P2PKH	SegWit P2SH-P2WPKH
Locking Script (scriptPubKey)	OP_DUP OP_HASH160 <pubKeyHash> OP_EQUALVERIFY OP_CHECKSIG	OP_HASH160 <redeemScriptHash> OP_EQUAL
scriptSig	Contains signature and public key	Contains redeem script (witness program)
Witness Field	Not used	Contains signature and public key
Signature Location	Stored inside scriptSig	Stored inside witness field
btcdeb Execution	Uses scriptSig and scriptPubKey	Uses witness stack and redeem script

In Legacy P2PKH transactions, the unlocking data (signature and public key) is stored directly in the **scriptSig** field. During script execution, the scriptSig pushes the signature and public key onto the stack, after which the locking script validates them.

In SegWit P2SH-P2WPKH transactions, the scriptSig contains only the **redeem script**, which represents the SegWit witness program. The actual signature and public key are stored separately in the **witness field**, which is processed during script validation.

This separation of signature data from the main transaction structure is the core design change introduced by SegWit.

8. Witness Discount Mechanism :

SegWit introduces a new method for calculating transaction size called **transaction weight**, which distinguishes between base transaction data and witness data.

The transaction weight is calculated using the following formula:

Weight = (Base Size × 4) + Witness Size

Where:

- **Base Size** represents the size of the transaction excluding witness data.
- **Witness Size** represents the size of the signature and witness data.

The **virtual size (vsize)** of a transaction is then calculated as:

vsize = Weight / 4

Because witness data is multiplied by a lower factor during weight calculation, it effectively receives a **75% discount compared to base transaction data**. This mechanism allows transactions using SegWit to have smaller virtual sizes even if their raw byte size is slightly larger.

9. Reason for Smaller Virtual Size in SegWit Transactions :

In Legacy transactions, all transaction data—including signatures and public keys—is stored within the main transaction structure. Therefore, the virtual size of the transaction is equal to the actual transaction size.

In SegWit transactions, signature and public key data are moved into the **witness section**. Since witness data is given a lower weight during the calculation of transaction weight, the resulting virtual size becomes smaller.

For example:

Legacy Transaction Size = 225 bytes

Virtual Size = 225 vbytes

SegWit Transaction Size = 247 bytes

Virtual Size = 166 vbytes

Although the SegWit transaction has a slightly larger raw size, its virtual size is significantly smaller due to the witness discount mechanism.

10. Benefits of SegWit :

SegWit introduces several improvements for both Bitcoin users and the overall network. By separating signature data from the main transaction structure, SegWit makes transactions more efficient and helps improve the scalability of the Bitcoin system.

- **User Benefits**

  One of the most noticeable advantages for users is **lower transaction fees**. Since transaction fees are calculated based on the virtual size (vsize) rather than the actual byte size, the witness discount reduces the effective size of SegWit transactions. This means users can send transactions at a lower cost compared to traditional Legacy transactions. SegWit transactions also make better use of block space, allowing more transactions to fit inside a block.

- **Network Benefits**

  For the Bitcoin network, SegWit increases the overall transaction capacity of each block. Because witness data is counted differently during weight calculation, blocks can include more transactions without increasing the block size limit in the traditional sense. This helps reduce congestion and improves block space efficiency, especially during periods of high transaction demand.

- **Protocol Benefits**

SegWit also solves the **transaction malleability problem**, where transaction IDs could previously be modified before confirmation. Fixing this issue improves transaction reliability and makes it easier to build more advanced protocols on top of Bitcoin. In addition, SegWit was designed to support future improvements to the Bitcoin protocol without requiring disruptive changes to the network.

11. SegWit and Layer-2 Scaling (Lightning Network) :

SegWit also plays an important role in enabling Layer-2 scaling solutions such as the **Lightning Network**.

The Lightning Network allows users to create **payment channels**, where multiple transactions occur off-chain between participants. Only the opening and closing transactions of the channel are recorded on the Bitcoin blockchain.

Before SegWit, transaction malleability made it difficult to implement reliable off-chain payment channels because transaction IDs could change after broadcasting. SegWit resolved this issue by separating signature data from the transaction ID calculation.

As a result, SegWit provides the necessary foundation for Layer-2 systems like the Lightning Network, which significantly improve Bitcoin scalability by enabling fast and low-cost transactions without overloading the blockchain.

12 . Conclusion :

In this experiment, we explored how Bitcoin transactions work by creating and analyzing both **Legacy P2PKH** and **SegWit P2SH-P2WPKH** transactions using the Bitcoin **Regtest network**. The Regtest environment allowed us to safely generate blocks and test transactions locally without using real Bitcoin.

In the first part of the experiment, Legacy P2PKH transactions were created between three addresses (A → B and B → C). The transactions were decoded using decoderawtransaction, allowing us to inspect important fields such as scriptPubKey, scriptSig, transaction size, virtual size, and weight. Using **btcdeb**, we stepped through the script execution process and observed how the unlocking script provides the signature and public key, while the locking script verifies ownership through hashing and signature validation.

In the second part, we created SegWit P2SH-P2WPKH transactions between SegWit addresses (A' → B' and B' → C'). Unlike Legacy transactions, the signature and public key were stored in the **witness field**, while the scriptSig contained only the redeem script. This design change separates signature data from the main transaction structure and reduces the effective transaction size.

When comparing both transaction types, we observed that Legacy transactions had a virtual size of **225 vbytes**, while SegWit transactions had a smaller virtual size of **166 vbytes**. This difference occurs because SegWit introduces a **witness discount**, meaning that witness data is counted with lower weight when calculating transaction size.

Overall, the experiment shows that SegWit improves transaction efficiency by reducing the effective transaction size, which results in **lower transaction fees and better utilization of block space**. In addition, SegWit fixes important protocol issues such as **transaction malleability** and enables advanced scaling solutions like the **Lightning Network**. These improvements make SegWit an important upgrade that helps enhance the scalability and reliability of the Bitcoin network.